

P01 VERSION HISTORY

P02 BLOCK DIAGRAM

P03 POWER TREE

P04 GPIO ASSIGNMENT

P05 POWER

P06 SOC1

P07 SOC2

P08 DDR3

P09 FLASH

P10 AUDIO

P11 USB CARD HDMI

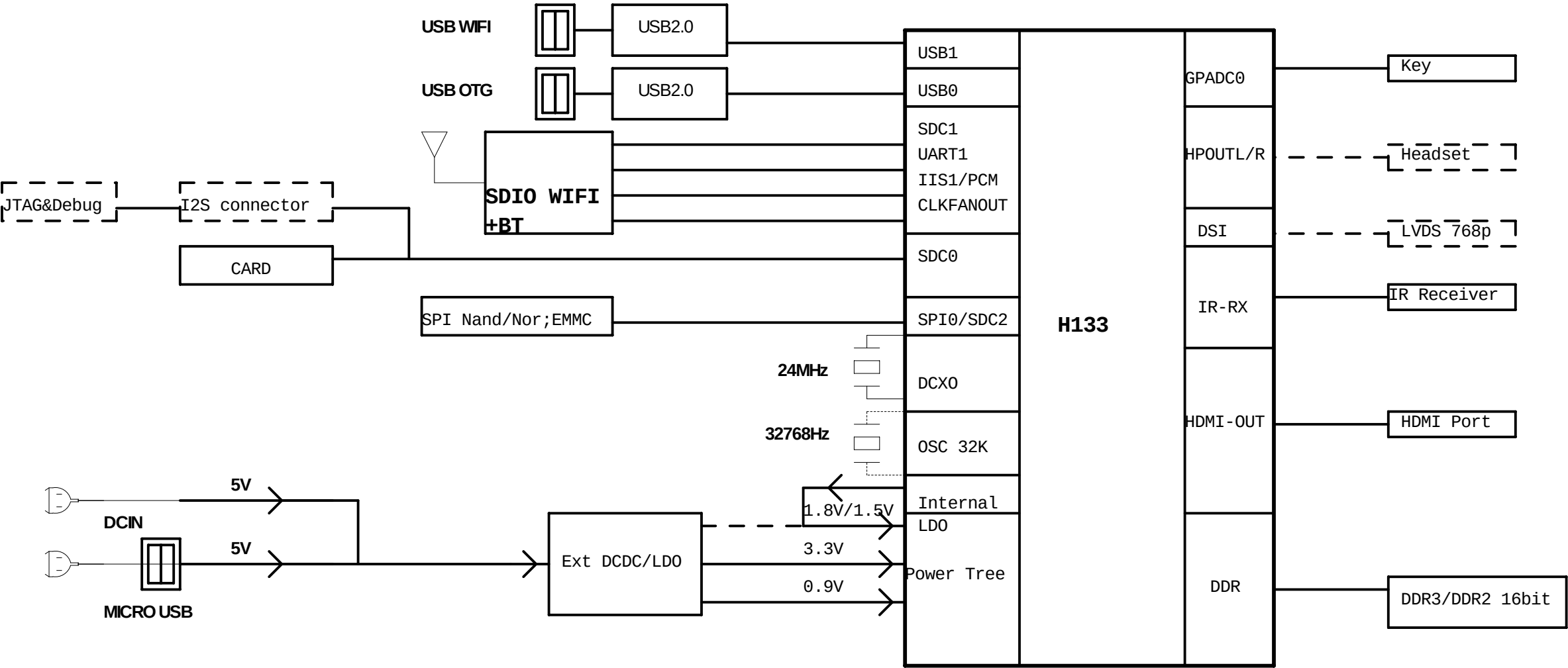
P12 WIFI BT

OPTIONAL

P13 DDR2

P14 USB WIFI

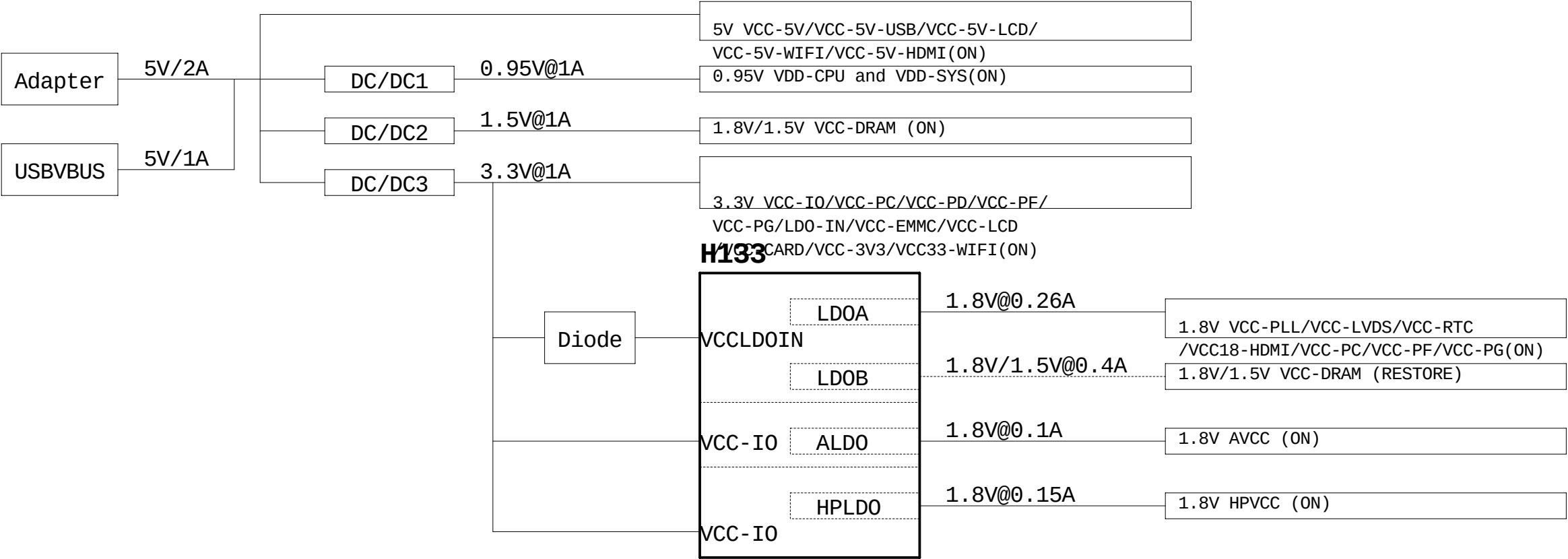
Ver 0.1	Releas version	2020-12-31			
Ver 1.0	AddAW859 Add UART2 /5 and SPI test point	2021-04-07	YJY	JIAYONG	YINWEI
Ver 1.1	ADD DDR2 ADD FALE FEL KEY ADD USB WIFI ADD DCDC4 for dram	2021-11-21	WJH	YJY	YINWEI
Ver 1.2	Merge VDD-CPU and vdd-SYS VCC-DRAM change to ext DCDC	2022-01-05	WJH	YJY	YINWEI



AllWinner Technology Co.,Ltd			
Design Name			
H133_DONGLE_STD			
Size	Page Name		Rev
B	02 BLOCK DIAGRAM		?
Date:	Friday, April 10, 2026	Sheet	2 of 14



DEFAULT POWER OFF



AllWinner Technology Co., Ltd

Design Name
H133_DONGLE_STD

Size B
Page Name
03 POWER TREE

Rev
?

Date: Friday, April 10, 2026 Sheet 3 of 14

D15	PB0	PWM3/IR_TX/TWI2_SCK/SPI1_WP/DBI_TE/UART0_TX/UART2_TX/SPDIF_OUT/PB_EINT0
D14	PB1	PWM4/I2S2_DOUT3/TWI2_SDA/I2S2_DIN3/UART0_RX/UART2_RX/IR_RX/PB_EINT1
D13	PB8	DMIC_DATA3/PWM5/TWI2_SCK/SPI1_HOLD/DBI_DCX/DBI_WRX/
C14	PB9	
C13	PB10	UART0_RX/UART1_RX/PB_EINT9 DMIC_DATA1/PWM7/TWI0_SCK/SPI1_MOSI/DBI_SDO/CLK_FANOUT0/
B15	PB11	UART1_RTS/PB_EINT10 DMIC_DATA0/PWM2/TWI0_SDA/SPI1_CLK/DBI_SCLK/CLK_FANOUT1/
B14	PB12	UART1_CLK/PWM0/SPDIF_IN/SPI1_CS/DBI_CSX/CLK_FANOUT2/IR_RX/PB_EINT12

Ball Number	Ball Name	GPIO Multiplex Function
F3	PC2	SPI0_CLK/SDC2_CLK/PC_EINT2
F2	PC3	SPI0_CS0/SDC2_CMD/PC_EINT3
F1	PC4	SPI0_MOSI/SDC2_D2/BOOT_SEL0/PC_EINT4
G3	PC5	SPI0_MISO/SDC2_D1/BOOT_SEL1/PC_EINT5
G2	PC6	SPI0_WP/SDC2_D0/UART3_TX/TWI3_SCK/DBG_CLK/PC_EINT6
H3	PC7	SPI0_HOLD/SDC2_D3/UART3_RX/TWI3_SDA/TCON_TRIG/PC_EINT7

Ball Number	Ball Name	GPIO Multiplex Function
N15	PD0	LCD0_D2/LVDS0_V0P/DSI_D0P/TWI0_SCK/PD_EINT0
N14	PD1	LCD0_D3/LVDS0_V0N/DSI_D0N/UART2_TX/PD_EINT1
M15	PD2	LCD0_D4/LVDS0_V1P/DSI_D1P/UART2_RX/PD_EINT2
M14	PD3	LCD0_D5/LVDS0_V1N/DSI_D1N/UART2_RTS/PD_EINT3
L15	PD4	LCD0_D6/LVDS0_V2P/DSI_CKP/UART2_CTS/PD_EINT4
L14	PD5	LCD0_D7/LVDS0_V2N/DSI_CKN/UART5_TX/PD_EINT5
K15	PD6	LCD0_D10/LVDS0_CKP/DSI_D2P/UART5_RX/PD_EINT6
K14	PD7	LCD0_D11/LVDS0_CKN/DSI_D2N/UART4_TX/PD_EINT7
J15	PD8	LCD0_D12/LVDS0_V3P/DSI_D3P/UART4_RX/PD_EINT8
J14	PD9	LCD0_D13/LVDS0_V3N/DSI_D3N/PWM6/PD_EINT9

B8	PG0	SDC1_CLK/UART3_TX/RGMII_RXCTRL/RMII_CRS_DV/PWM7/PG_EINT0
C9	PG1	SDC1_CMD/UART3_RX/RGMII_RXD0/RMII_RXD0/PWM6/PG_EINT1
A8	PG2	SDC1_D0/UART3_RTS/RGMII_RXD1/RMII_RXD1/UART4_TX/PG_EINT2
B7	PG3	SDC1_D1/UART3_CTS/RGMII_TXCK/RMII_TXCK/UART4_RX/PG_EINT3
A6	PG4	SDC1_D2/UART5_TX/RGMII_TXD0/RMII_TXD0/PWM5/PG_EINT4
C7	PG5	SDC1_D3/UART5_RX/RGMII_TXD1/RMII_TXD1/PWM4/PG_EINT5
B4	PG6	UART1_TX/TWI2_SCK/RGMII_TXD2/PWM1/PG_EINT6
A3	PG7	UART1_RX/TWI2_SDA/RGMII_TXD3/SPDIF_IN/PG_EINT7
B3	PG8	UART1_RTS/TWI1_SCK/RGMII_RXD2/UART3_TX/PG_EINT8
A2	PG9	UART1_CTS/TWI1_SDA/RGMII_RXD3/UART3_RX/PG_EINT9
C4	PG10	PWM3/TWI3_SCK/RGMII_RXCK/CLK_FANOUT0/IR_RX/PG_EINT10
B6	PG11	I2S1_MCLK/TWI3_SDA/EPHY_25M/CLK_FANOUT1/TCON_TRIG/PG_EINT11
C6	PG12	I2S1_LRCK/TWI0_SCK/RGMII_TXCTRL/RMII_TXEN/CLK_FANOUT2/PWM0/UART1_TX/PG_EINT12
B5	PG13	I2S1_BCLK/TWI0_SDA/RGMII_CLKIN/RMII_RXER/PWM2/LEDC_D0/UART1_RX/PG_EINT13
C5	PG14	I2S1_DIN0/TWI2_SCK/MDC/I2S1_DOUT1/SPI0_WP/UART1_RTS/PG_EINT14
A4	PG15	I2S1_DOUT0/TWI2_SDA/MDIO/I2S1_DIN1/SPI0_HOLD/UART1_CTS/PG_EINT15
B2	PG16	IR_RX/TCON_TRIG/PWM5/CLK_FANOUT2/SPDIF_IN/LEDC_D0/PG_EINT16
C10	PG17	UART2_TX/TWI3_SCK/PWM7/CLK_FANOUT0/IR_TX/UART0_TX/PG_EINT17
B9	PG18	UART2_RX/TWI3_SDA/PWM6/CLK_FANOUT1/SPDIF_OUT/UART0_RX/PG_EINT18

Ball Number	Ball Name	GPIO Multiplex Function
B1	PF0	SDC0_D1/JTAG_MS/R_JTAG_MS/I2S2_DOUT1/I2S2_DIN0/PF_EINT0
C3	PF1	SDC0_D0/JTAG_DI/R_JTAG_DI/I2S2_DOUT0/I2S2_DIN1/PF_EINT1
C2	PF2	SDC0_CLK/UART0_TX/TWI0_SCK/LEDC_D0/SPDIF_IN/PF_EINT2
D3	PF3	SDC0_CMD/JTAG_D0/R_JTAG_D0/I2S2_BCLK/PF_EINT3
D2	PF4	SDC0_D3/UART0_RX/TWI0_SDA/PWM6/IR_TX/PF_EINT4
D1	PF5	SDC0_D2/JTAG_CK/R_JTAG_CK/I2S2_LRCK/PF_EINT5
E2	PF6	SPDIF_OUT/IR_RX/I2S2_MCLK/PWM5/PF_EINT6

	AllWinner Technology Co.,Ltd		
	Design Name		
	H133_DONGLE_STD		
	Size B	Page Name 04 GPIO ASSIGNMENT	Rev ?
Date: Friday, April 10, 2026		Sheet 4	of 14

PC

H133-BGA196
BGA196P65B1000_1000H117
U11-B

PB

H133-BGA196
BGA196P65B1000_1000H117
U11-D

PD

H133-BGA196
BGA196P65B1000_1000H117
U11-F

PF

H133-BGA196
BGA196P65B1000_1000H117

U11-G

PG

H133-BGA196
BGA196P65B1000_1000H117

PC3/SPI0_CS0/SDC2_CMD/PC_EINT3
PC4/SPI0_MOSI/SDC2_D2/BOOT_SEL0/PC_EINT4
PC5/SPI0_MISO/SDC2_D1/BOOT_SEL1/PC_EINT5
PC6/SPI0_WP/SDC2_D0/UART3_TX/TW3_SCK/DB
PC7/SPI0_HOLD/SDC2_D3/UART3_RX/TW3_SDA/
VCC_PC

F1
G3
G2
G2
H3
E5

MMC-PC

SPI0-CS0
SPI0-MOSI
SPI0-MISO
SPI0-WP
SPI0-HOLD

3.3V

D15
D14
D13
C14
C13
B15
B14
F11

CPU-P-PWM
IR-RX
UART0-TX
UART0-RX
LCD-RST
LCD-PWM

VCC-IO

3.3V

N15
N14
M15
M14
L15
L14
K15
K14
J15
J14
G11
J11

LVDS-V0P
LVDS-V0N
LVDS-V1P
LVDS-V1N
LVDS-V2P
LVDS-V2N
LVDS-CKP
LVDS-CKN
LVDS-V3P
LVDS-V3N

DST-D0P
LVDS-V0N
LVDS-V1P
LVDS-V1N
LVDS-V2P
LVDS-V2N
LVDS-CKP
LVDS-CKN
LVDS-V3P
LVDS-V3N

VCC-LVDS
VCC-PD

3.3V

B1
C3
C2
D3
D2
D1
E2
E5

SDC0-D1
SDC0-D0
SDC0-CLK
SDC0-CMD
SDC0-D3
SDC0-D2
SDC0-DET

SDC0-D1
SDC0-D0
SDC0-CLK
SDC0-CMD
SDC0-D3
SDC0-D2
SDC0-DET

VCC-PF

1.8V / 3.3V

B8
C9
A8
B7
A6
C7
B4
A3
B3
A2
C4
B6
C6
B5
C5
A4
B2
C10
B9
D5

UART_TX
UART_RX
CTP-RST
CTP-INT
LED-PWM
TW2-SCK
TW2-SDA
CTP-SCK
CTP-SDA
I2S1-MCLK
I2S1-LRLK
I2S1-BCLK
I2S1-DIN0
LCD-BL-EN
SPDIF-OUT

15
15
[13]
14
14
15
15
14
14
14
14
[18]

VCC-PG

CF1
100PF
C0402

GND

CS1
0.1uF
C0402

GND

VCC-IO

CS2
1uF
C0402

CS3
0.1uF
C0402

GND

VCC-PD

CS4
0.1uF
C0402

GND

VCC-LVDS

CS5
0.1uF
C0402

GND

VCC-PF

CS6
0.1uF
C0402

GND

VCC-PG

CS7
0.1uF
C0402

GND

AllWinner Technology Co.,Ltd

Design Name
H133_DONGLE_STD

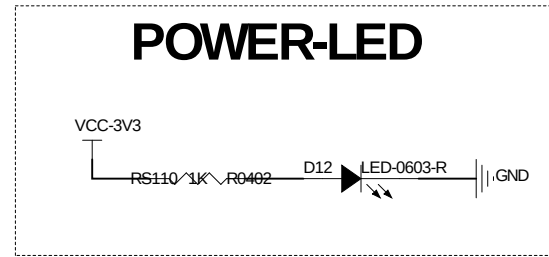
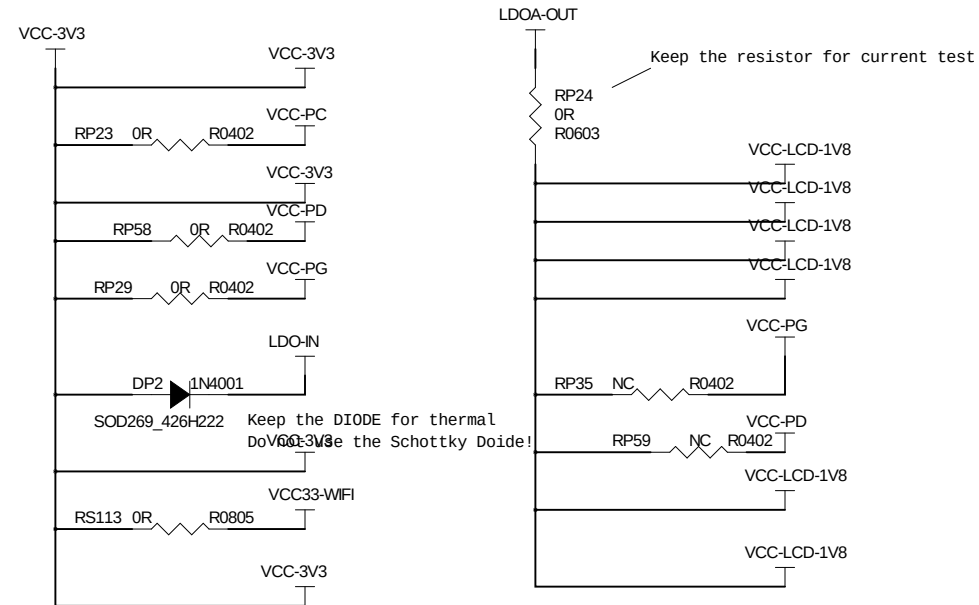
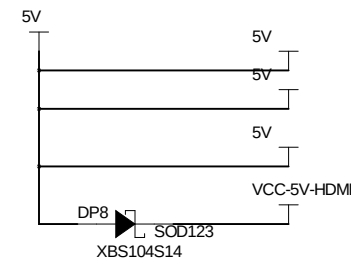
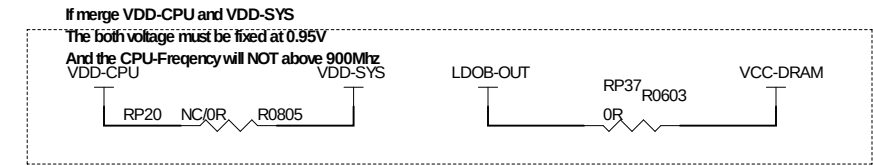
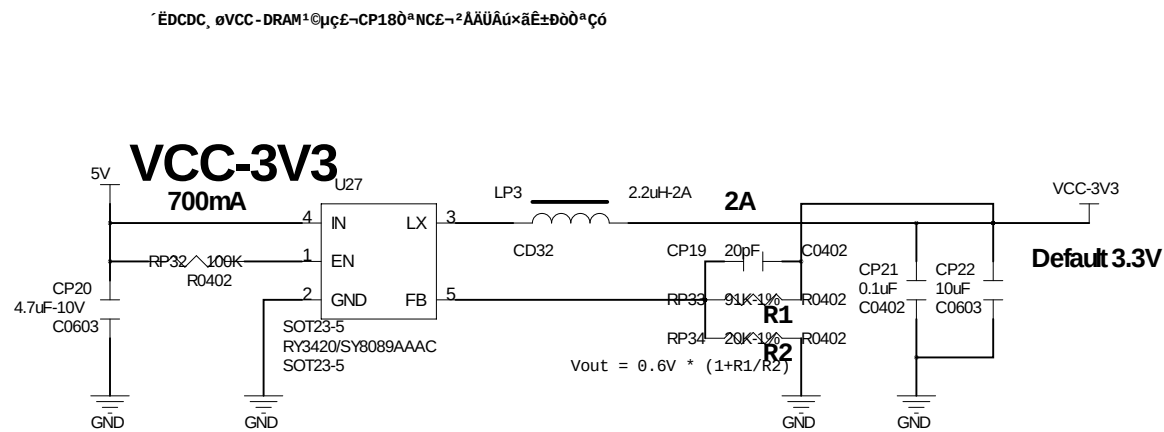
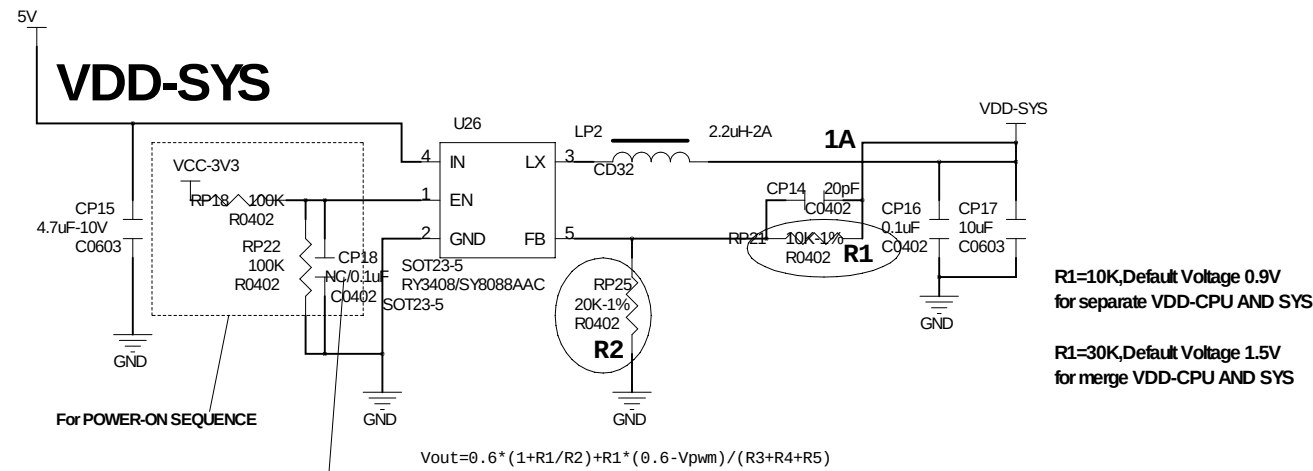
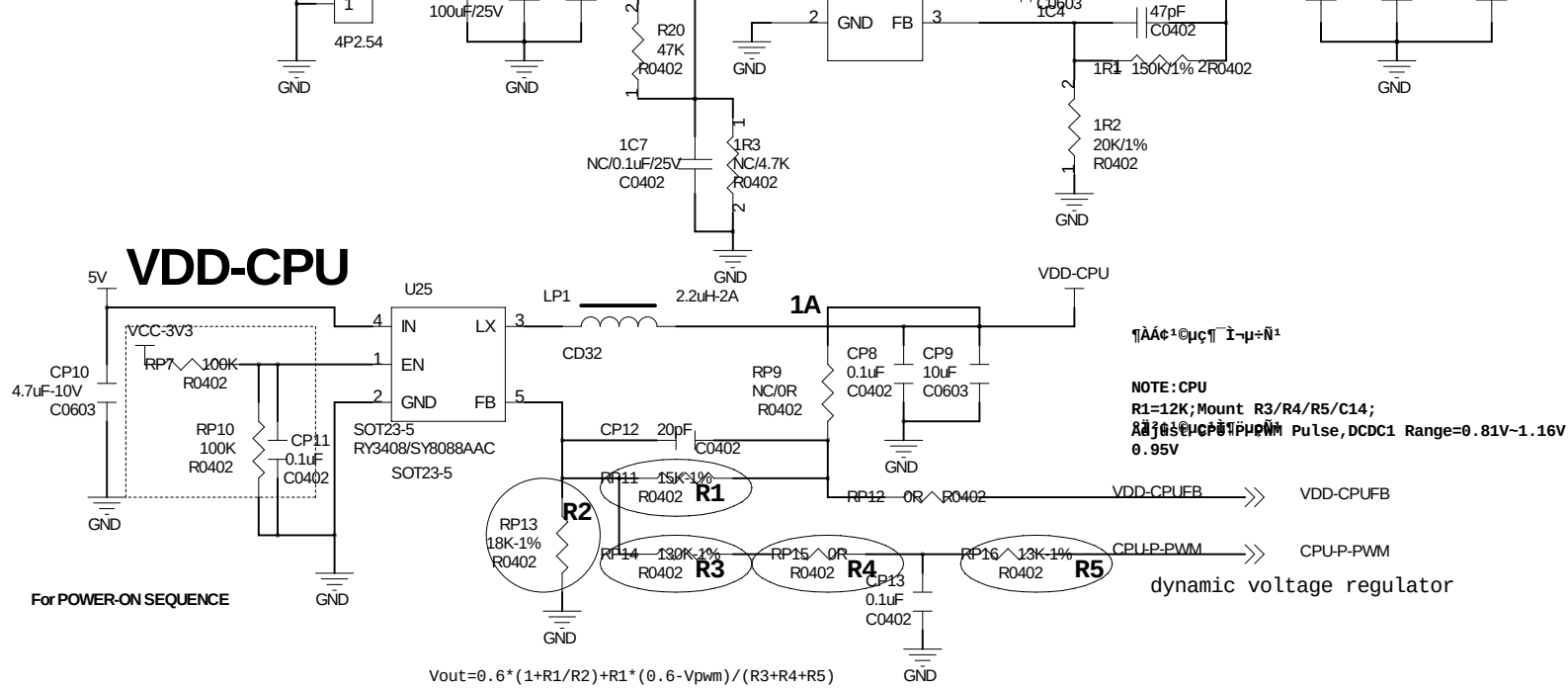
Size
B

Page Name
07 SOC1

Rev
?

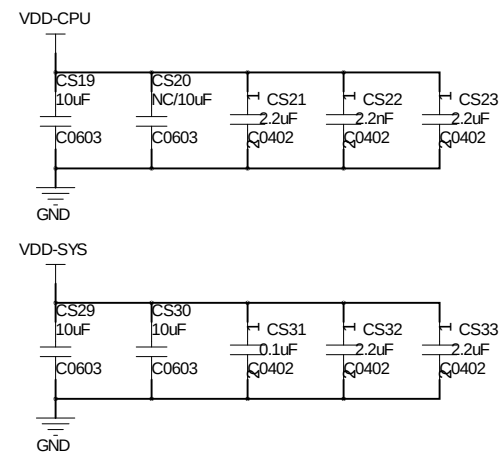
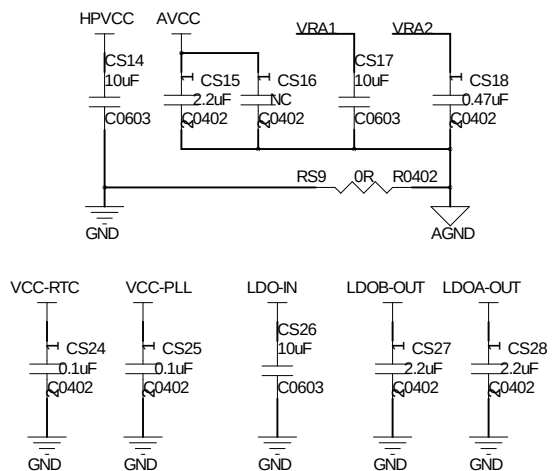
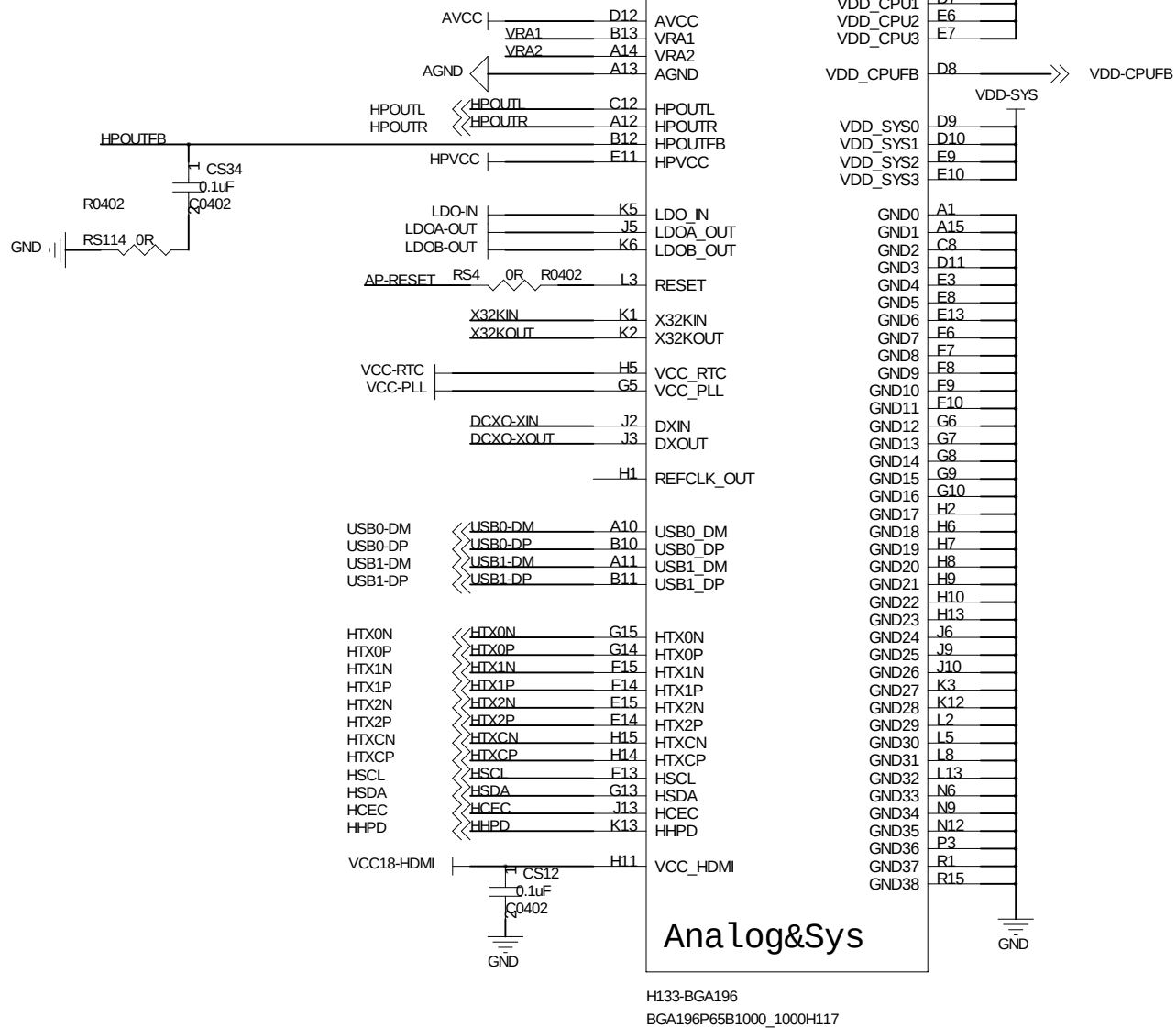
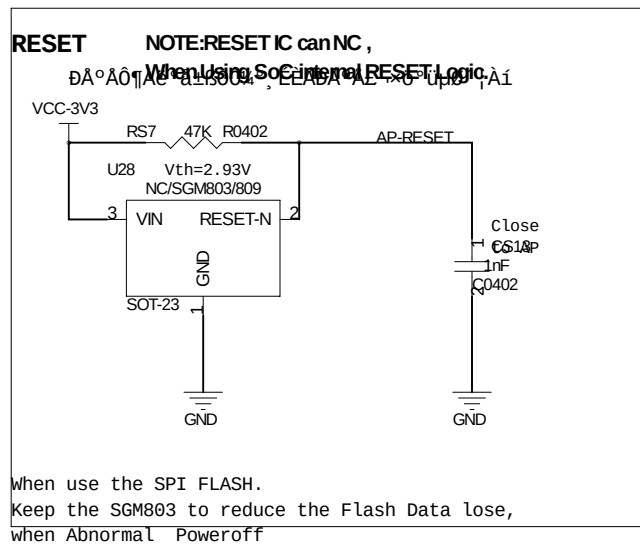
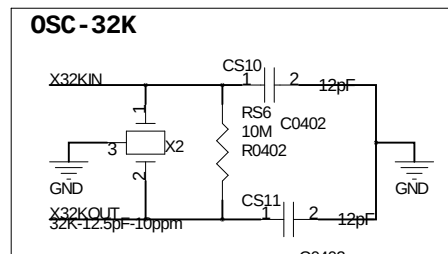
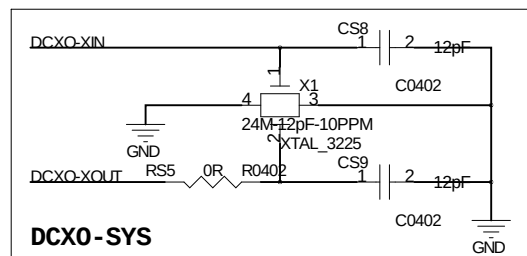
Date: Friday, April 10, 2026

Sheet 5 of 14

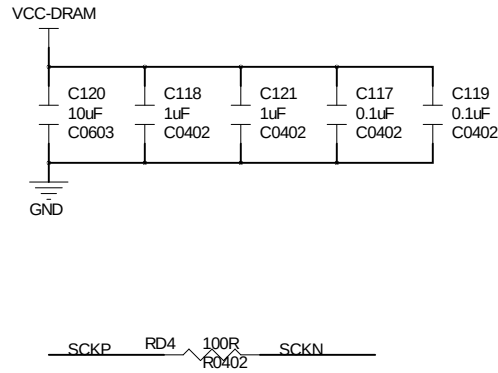
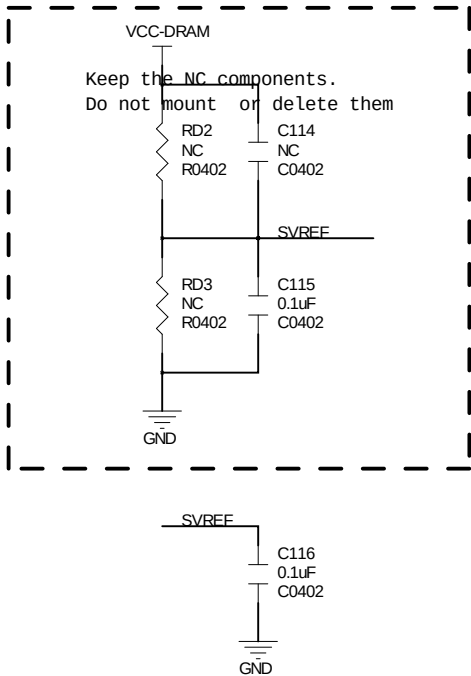
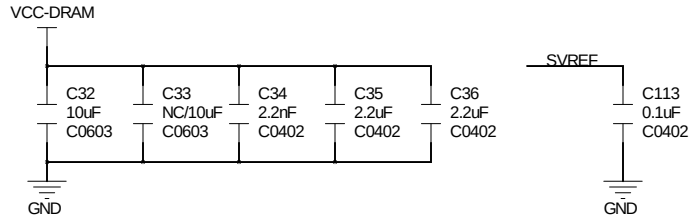
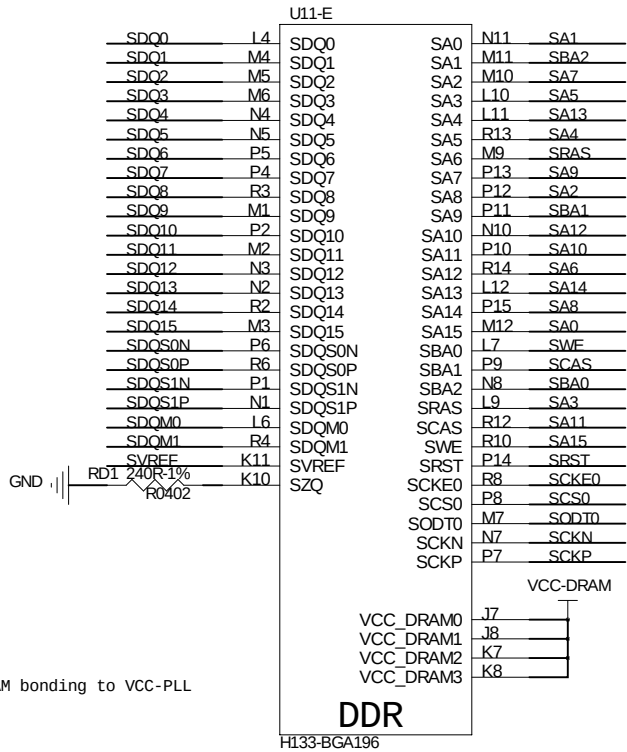


	DCDC1 (EXT)	DCDC2 (EXT)	DCDC3 (EXT)	DCDC4 (EXT)	LDOB (IN)	LDOA (IN)	REMARK
Default PLAN	VDD-CPU VDD-SYS (0.95V)	VCC-DRAM 1.5/1.8V	VCC-WIFI VCC-3V3 LDOIN	NC	NC	VCC-PLL VCC-RTC VCC-1V8	Merge VDD-CPU and VDD-SYS ,that Cpu-frequency below 900Mhz.The Power consume is below2.5W@4K30,And the thermal will be very low.
Performance PLAN	VDD-CPU (0.8~ 1.16V)	VDD-SYS 0.9V	VCC-WIFI VCC-3V3 LDOIN	VCC-DRAM 1.5/1.8V	NC	VCC-PLL VCC-RTC VCC-1V8	Separate VDD-CPU and VDD-SYS ,The Cpu-frequency upon to 1.2Ghz .The Power consume is below2.5W@4K30,And the thermal will be very low.
Econormy PLAN	VDD-CPU VDD-SYS (0.95V)	NC	VCC-WIFI VCC-3V3 LDOIN	NC	VCC-DRAM 1.5/1.8V	VCC-PLL VCC-RTC VCC-1V8	Merge VDD-CPU and VDD-SYS ,that Cpu-frequency below 900Mhz.VCC-Dram be supplied by internal LDOB,instead of Ext DCDC.

AllWinner Technology Co.,Ltd			
Design Name			
H133_DONGLE_STD			
Size B	Page Name		Rev ?
	05 POWER		
Date:	Friday, April 10, 2026	Sheet	6 of 14

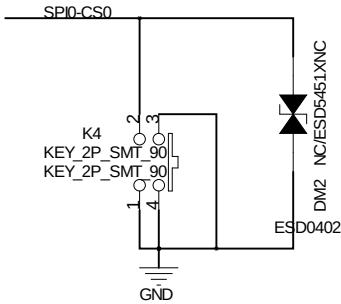
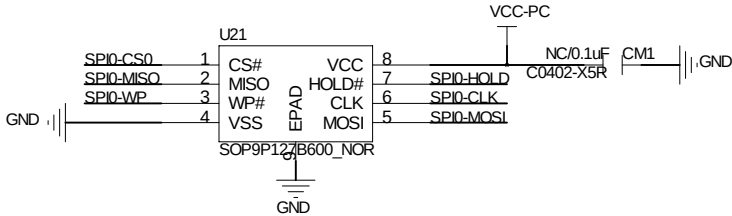
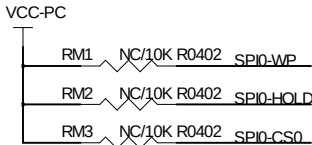
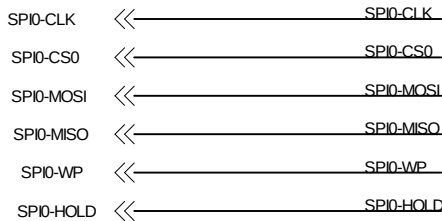


NOTE:
1.VDD18-DRAM bonding to VCC-PLL



AllWinner Technology Co.,Ltd			
Design Name			
H133_DONGLE_STD			
Size	Page Name		Rev
B	09 DDR3		?
Date:	Friday, April 10, 2026		Sheet 8 of 14

SPI



UBOOT

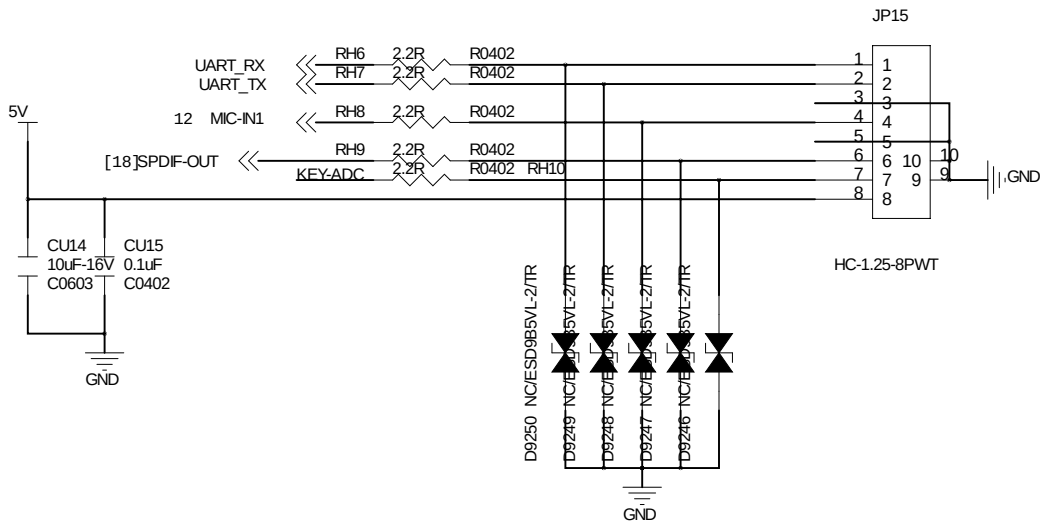
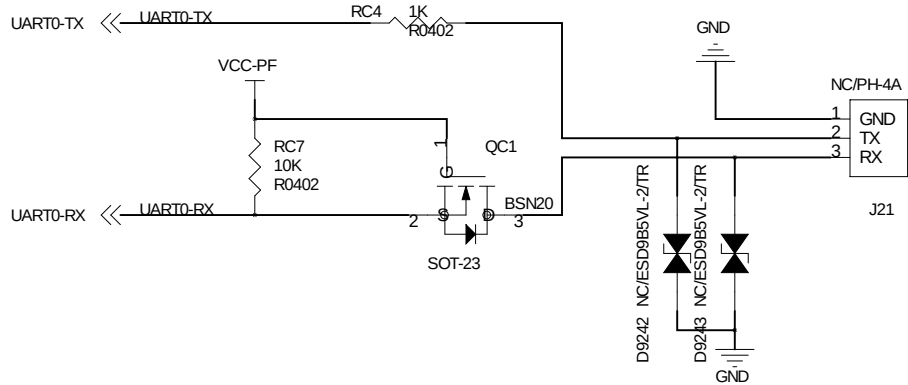
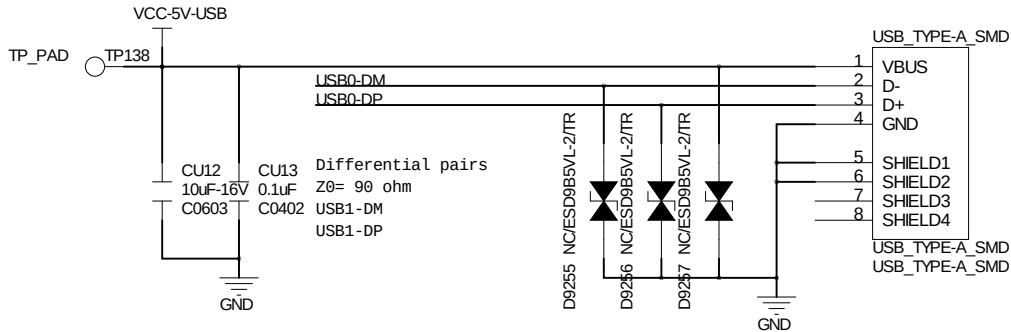
AllWinner Technology Co.,Ltd

Design Name H133_DONGLE_STD

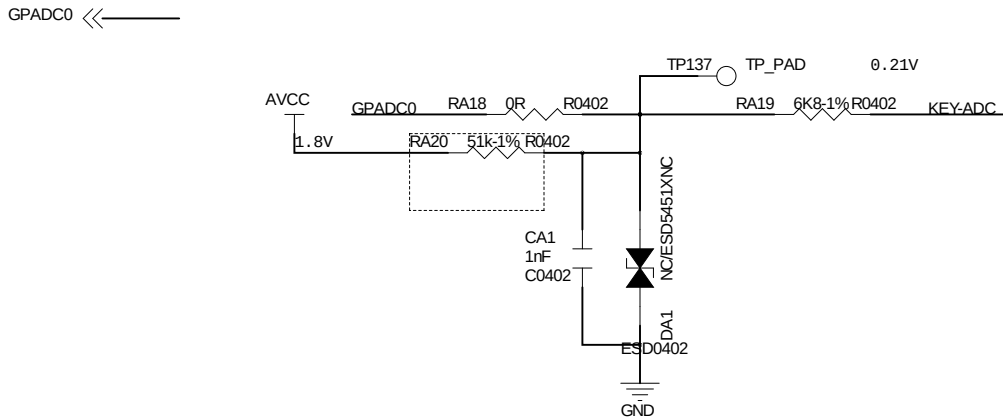
Size B Page Name 10 FLASH Rev ?

Date: Friday, April 10, 2026 Sheet 9 of 14

USB0-DM
USB0-DP

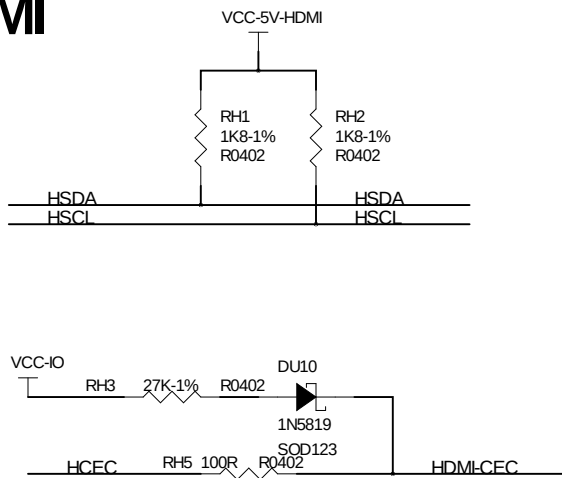


KEY ADC

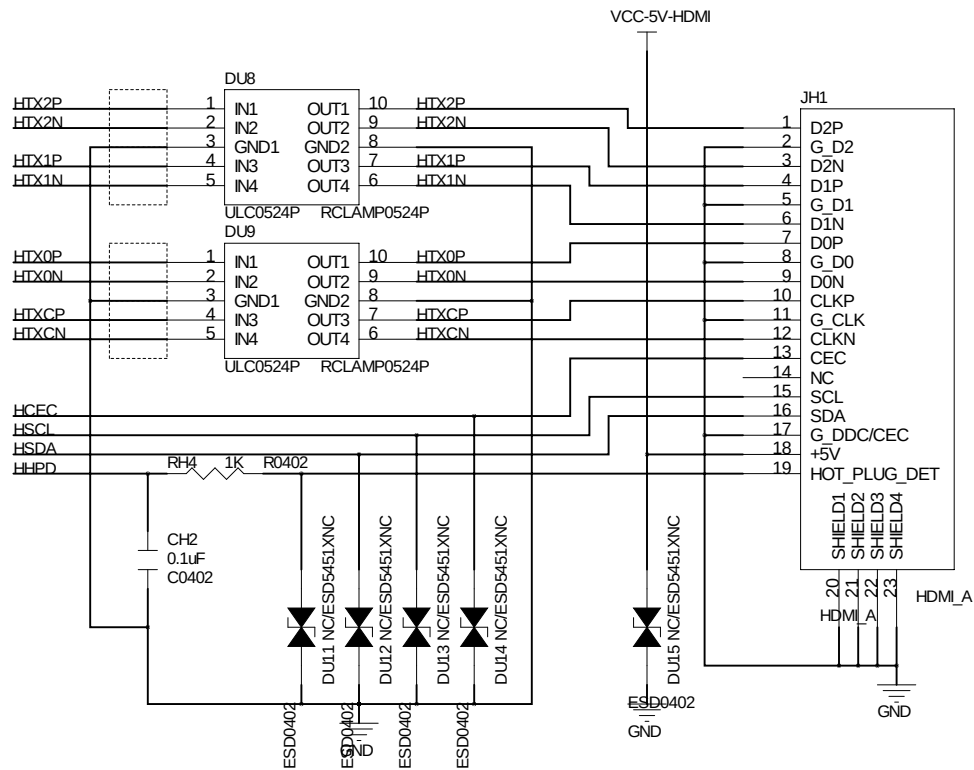


HCEC
HHPD
HSCL
HSDA
HTX0N
HTX0P
HTX1N
HTX1P
HTX2N
HTX2P
HTXCN
HTXCP

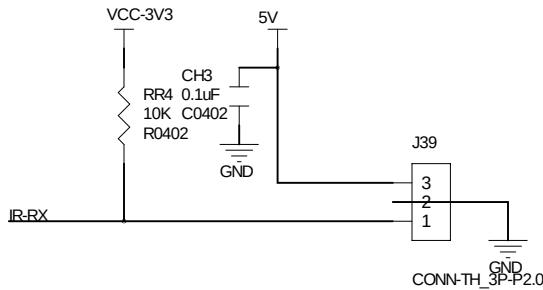
HDMI



NOTE:
Differential pairs
Z0= 100 ohm



IR-RX



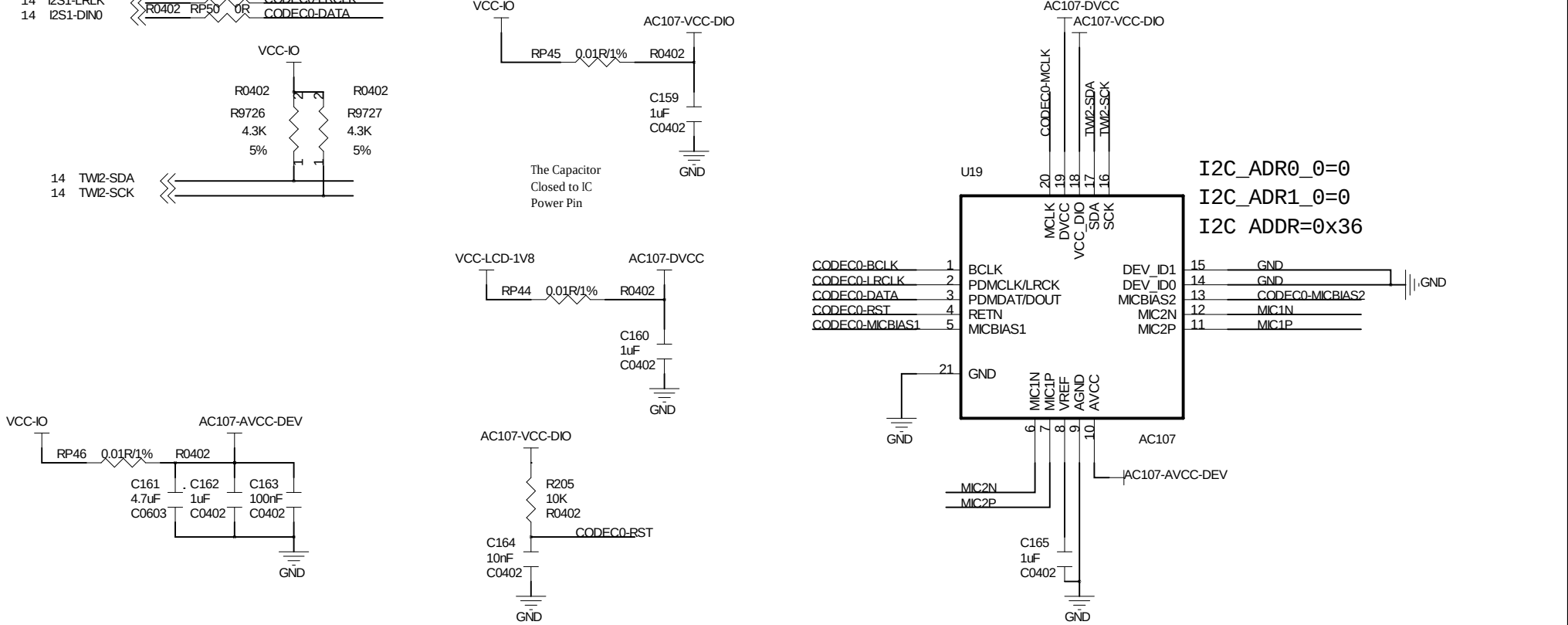
AllWinner Technology Co., Ltd

Design Name
H133_DONGLE_STD

Size B Page Name
13 USB CARD HDMI

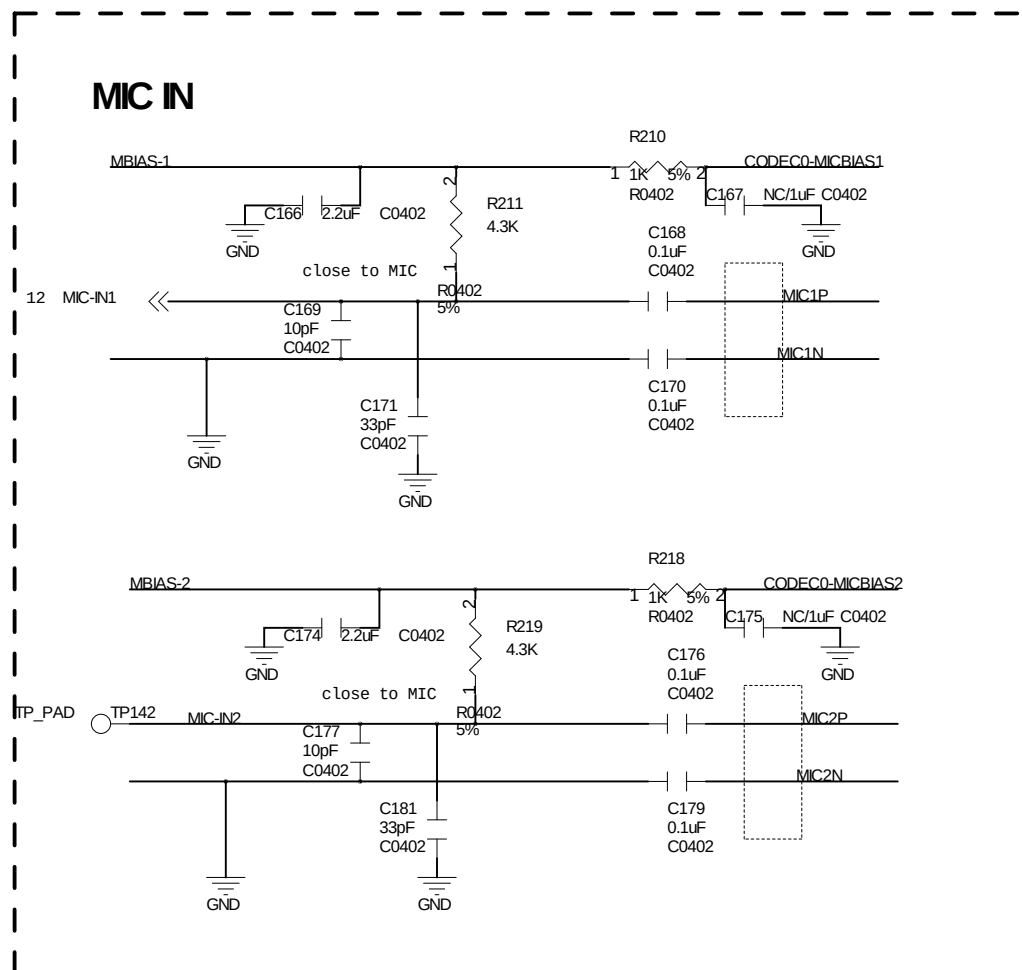
Rev
?

Date: Friday, April 10, 2026 Sheet 10 of 14



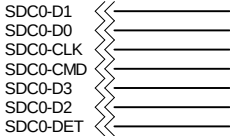
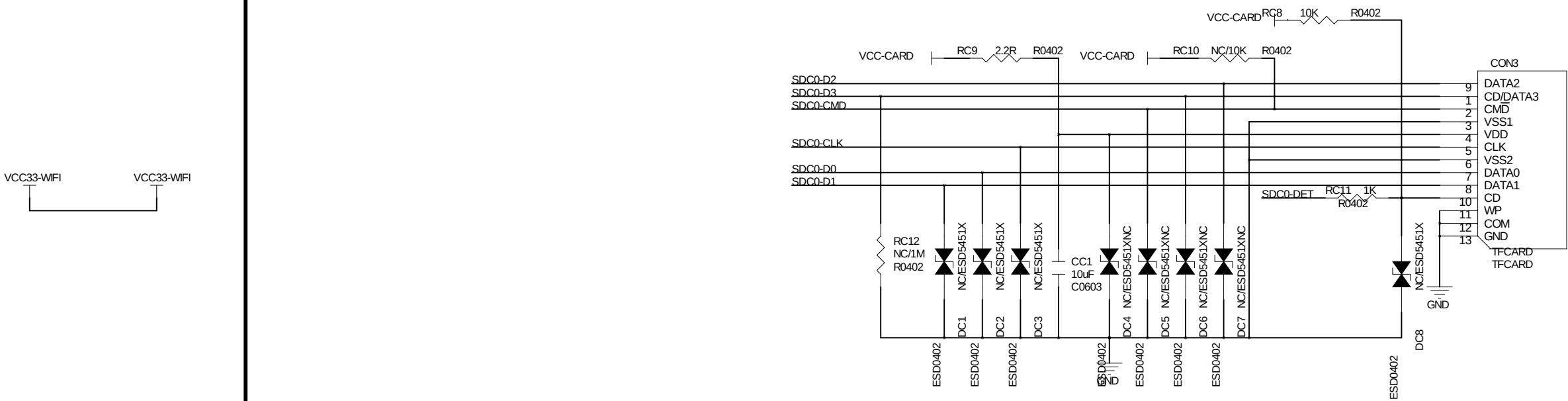
HPOUTL
HPOUTR

Audio

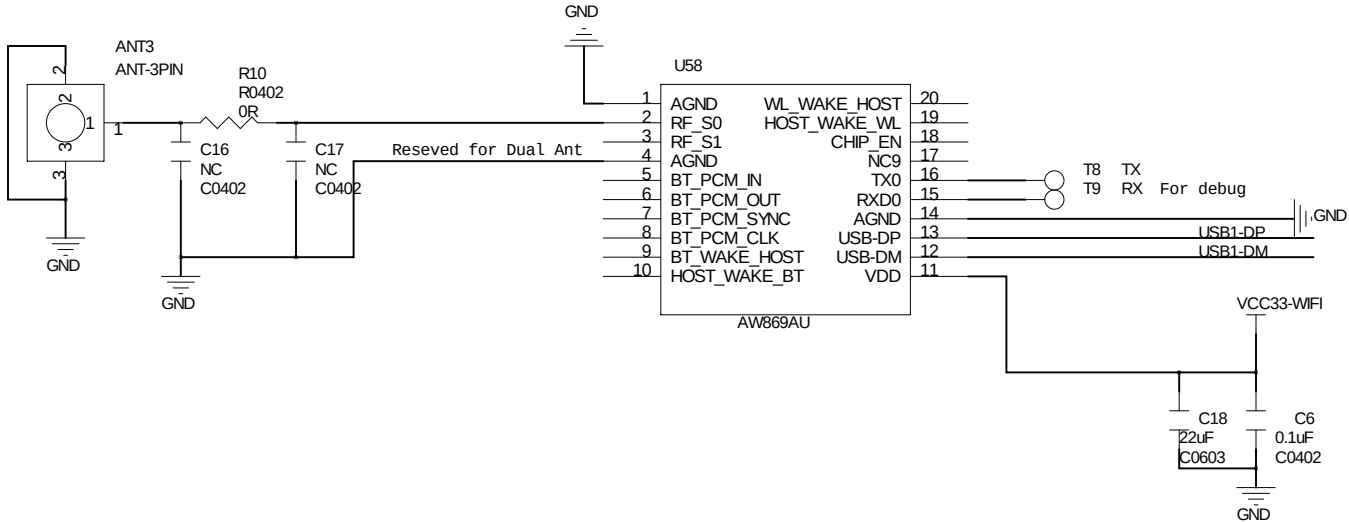


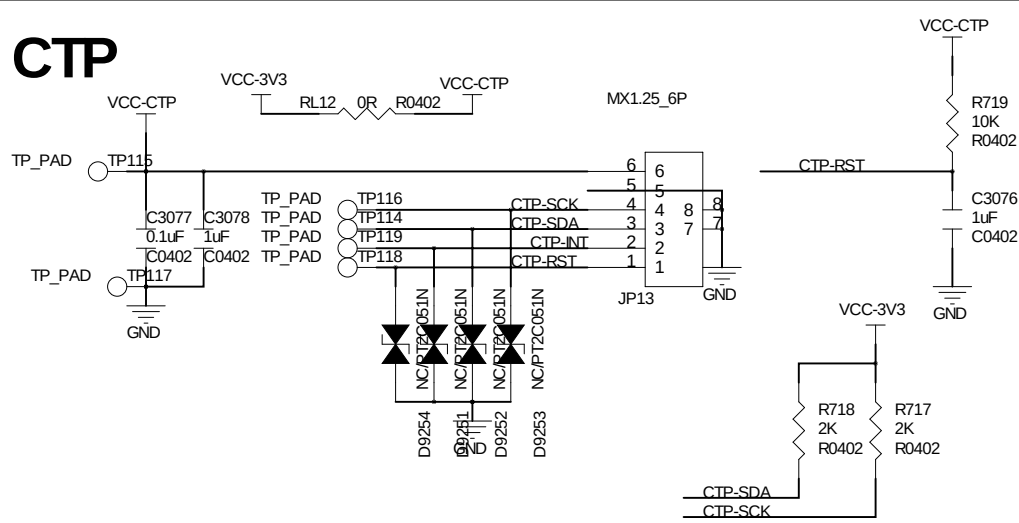
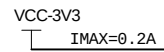
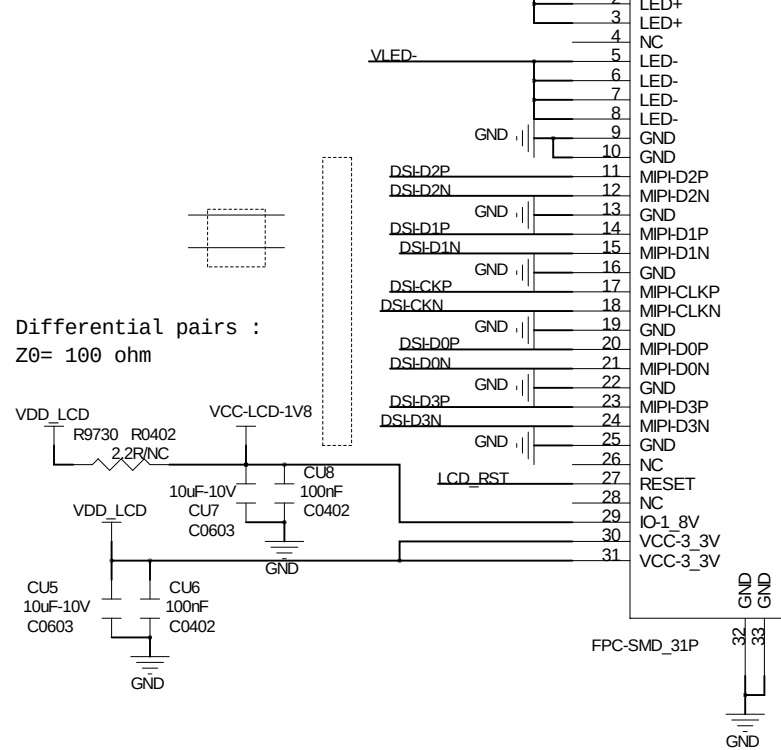
Title		
Size B	Document Number	Rev ?
Date:	Friday, April 10, 2026	Sheet 11 of 14

SDC0/JTAG Debug

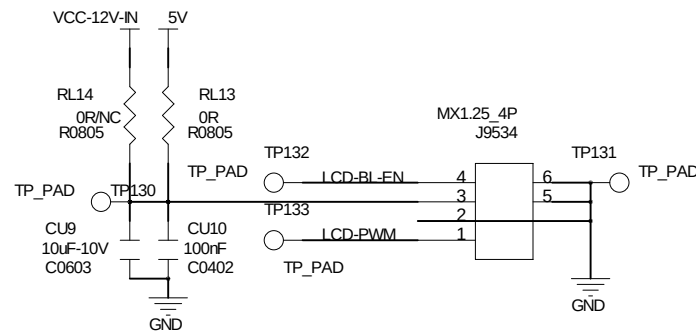


USB WIFI

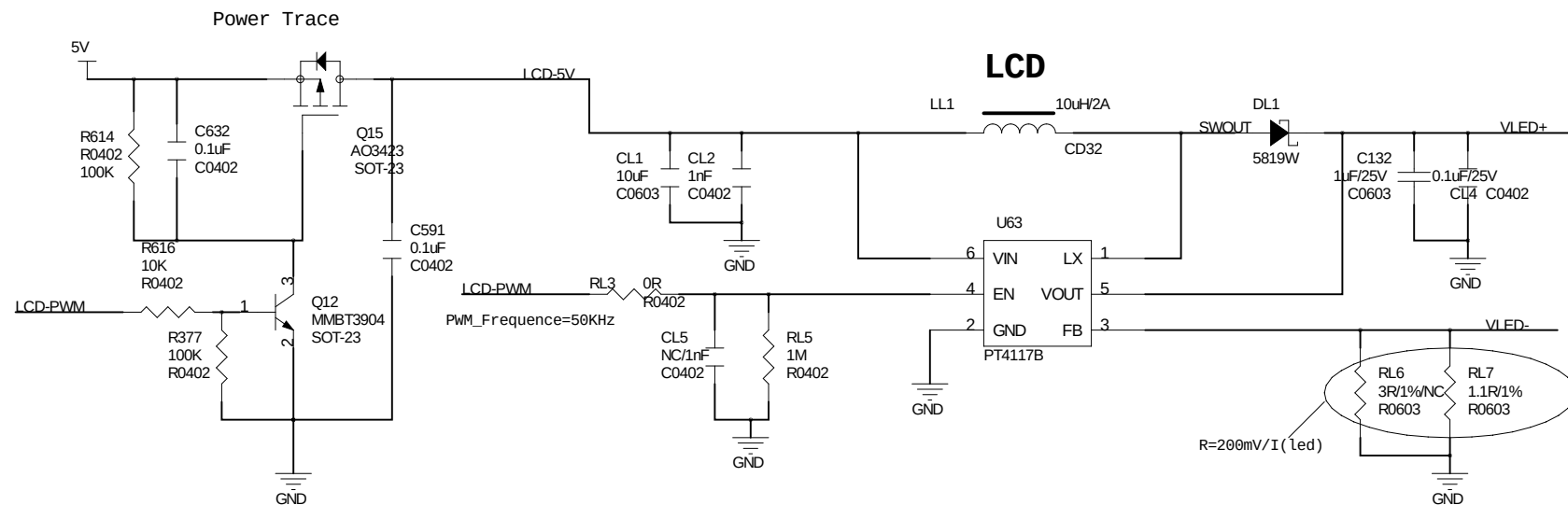
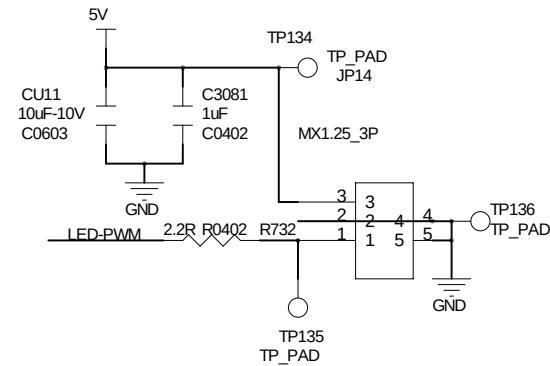




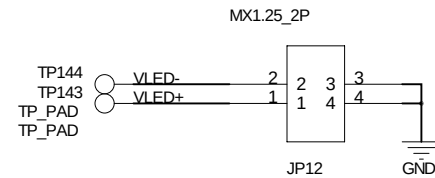
BACKLIGHT



氛围灯



BACKLIGHT



AllWinner Technology Co.,Ltd				
Design Name A100_A133_B3_AXP717_LPDDR4				
Size B	Page Name 13 LCM			Rev ?
Date:	Friday, April 10, 2026	Sheet	14	of 14